

Creation Date 22-Sep-2009

Revision Date 25-Jan-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                       |
|---------------------------|---------------------------------------|
| Product Description:      | <b>Borane-tert-butylamine complex</b> |
| Cat No. :                 | <b>L13177</b>                         |
| Synonyms                  | tert-Butylamineborane                 |
| CAS No                    | 7337-45-3                             |
| Molecular Formula         | C <sub>4</sub> H <sub>14</sub> B N    |
| REACH registration number | -                                     |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Flammable solids

Category 1 (H228)

### Health hazards

Acute oral toxicity

Category 3 (H301)

Acute dermal toxicity

Category 3 (H311)

Acute Inhalation Toxicity - Dusts and Mists

Category 4 (H332)

Skin Corrosion/Irritation

Category 1 A (H314)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

### Environmental hazards

Chronic aquatic toxicity

Category 2 (H411)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

H228 - Flammable solid

H332 - Harmful if inhaled

H314 - Causes severe skin burns and eye damage

H411 - Toxic to aquatic life with long lasting effects

H301 + H311 - Toxic if swallowed or in contact with skin

### Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

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| Component                                        | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                            |
|--------------------------------------------------|-----------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | 7337-45-3 | EEC No. 230-851-5 | 95       | Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Flam. Sol. 1 (H228)<br>Aquatic Chronic 2 (H411) |
| tert-Butylamine                                  | 75-64-9   | EEC No. 200-888-1 | <5       | Flam Liq. 2 (H225)<br>Acute Tox. 4 (H302)<br>Skin Corr. 1A (H314)<br>Eye Dam. 1 (H318)<br>Acute Tox. 3 (H331)<br>Aquatic Chronic 3 (H412)    |

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                     |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                                                 |
|--------------------|-------------------------------------------------|
| Notes to Physician | Treat symptomatically. Symptoms may be delayed. |
|--------------------|-------------------------------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

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## Extinguishing media which must not be used for safety reasons

Do not use halon type extinguisher.

## 5.2. Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes.

## Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen, Oxides of boron.

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid dust formation.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Avoid dust formation.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep under nitrogen. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 4.1B

**Switzerland - Storage of hazardous substances**

Storage class - SC 4.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

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## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s):

| Component       | European Union | The United Kingdom | France | Belgium | Spain |
|-----------------|----------------|--------------------|--------|---------|-------|
| tert-Butylamine |                |                    |        |         | Piel  |

| Component       | Italy | Germany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Portugal | The Netherlands | Finland |
|-----------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|---------|
| tert-Butylamine |       | <p>TWA: 2 ppm (8 Stunden). AGW - ceiling factor 2.5; exposure factor 2</p> <p>TWA: 6.1 mg/m<sup>3</sup> (8 Stunden). AGW - ceiling factor 2.5; exposure factor 2</p> <p>TWA: 2 ppm (8 Stunden). MAK even if the MAK value is adhered to, "odor-associated" symptoms cannot be ruled out in individual cases</p> <p>TWA: 6.1 mg/m<sup>3</sup> (8 Stunden). MAK even if the MAK value is adhered to, "odor-associated" symptoms cannot be ruled out in individual cases</p> <p>Höhepunkt: 4 ppm</p> <p>Höhepunkt: 12.2 mg/m<sup>3</sup></p> |          |                 |         |

| Component       | Austria                                                                                                                                                          | Denmark                                                              | Switzerland | Poland | Norway                                                               |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------|--------|----------------------------------------------------------------------|
| tert-Butylamine | <p>MAK-KZGW: 25 ppm 15 Minuten</p> <p>MAK-KZGW: 75 mg/m<sup>3</sup> 15 Minuten</p> <p>MAK-TMW: 5 ppm 8 Stunden</p> <p>MAK-TMW: 15 mg/m<sup>3</sup> 8 Stunden</p> | <p>Ceiling: 5 ppm</p> <p>Ceiling: 15 mg/m<sup>3</sup></p> <p>Hud</p> |             |        | <p>Hud</p> <p>Ceiling: 5 ppm</p> <p>Ceiling: 15 mg/m<sup>3</sup></p> |

| Component       | Estonia | Gibraltar | Greece                                                                                                | Hungary | Iceland                                                                  |
|-----------------|---------|-----------|-------------------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------|
| tert-Butylamine |         |           | <p>STEL: 5 ppm</p> <p>STEL: 15 mg/m<sup>3</sup></p> <p>TWA: 5 ppm</p> <p>TWA: 15 mg/m<sup>3</sup></p> |         | <p>STEL: 5 ppm</p> <p>STEL: 15 mg/m<sup>3</sup></p> <p>Skin notation</p> |

| Component       | Russia | Slovak Republic | Slovenia                                                                                                                                     | Sweden                                                                                               | Turkey |
|-----------------|--------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|
| tert-Butylamine |        |                 | <p>TWA: 6.1 mg/m<sup>3</sup> 8 urah</p> <p>TWA: 2 ppm 8 urah</p> <p>STEL: 4 ppm 15 minutah</p> <p>STEL: 12.2 mg/m<sup>3</sup> 15 minutah</p> | <p>Binding STEL: 5 ppm 15 minuter</p> <p>Binding STEL: 15 mg/m<sup>3</sup> 15 minuter</p> <p>Hud</p> |        |

#### Biological limit values

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This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                         | Fresh water      | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)          |
|-----------------------------------|------------------|-------------------------------|--------------------|------------------------------------|-----------------------------|
| tert-Butylamine<br>75-64-9 ( <5 ) | PNEC = 0.016mg/L | PNEC = 39mg/kg<br>sediment dw | PNEC = 0.16mg/L    | PNEC = 72mg/L                      | PNEC = 7.76mg/kg<br>soil dw |

| Component                         | Marine water         | Marine water sediment          | Marine water Intermittent | Food chain | Air |
|-----------------------------------|----------------------|--------------------------------|---------------------------|------------|-----|
| tert-Butylamine<br>75-64-9 ( <5 ) | PNEC =<br>0.0016mg/L | PNEC = 3.9mg/kg<br>sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material              | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|-----------------------------|--------------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

A NIOSH/MSHA approved air purifying dust or mist respirator or European Standard EN 149.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

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|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | <b>Recommended Filter type:</b> low boiling organic solvent Type AX Brown conforming to EN371 Type A Brown                                                                                                                                                                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                                                                                                                                                                                                                             |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                |                                          |
|------------------------------------------------|--------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Powder Solid                   |                                          |
| <b>Appearance</b>                              | Off-white                      |                                          |
| <b>Odor</b>                                    | No information available       |                                          |
| <b>Odor Threshold</b>                          | No data available              |                                          |
| <b>Melting Point/Range</b>                     | 96 - 101 °C / 204.8 - 213.8 °F |                                          |
| <b>Softening Point</b>                         | No data available              |                                          |
| <b>Boiling Point/Range</b>                     | No information available       |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable                 | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available       |                                          |
| <b>Explosion Limits</b>                        | No data available              |                                          |
| <b>Flash Point</b>                             | No information available       | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 177 °C / 350.6 °F              |                                          |
| <b>Decomposition Temperature</b>               | 71 °C                          |                                          |
| <b>pH</b>                                      | Not applicable                 |                                          |
| <b>Viscosity</b>                               | Not applicable                 | Solid                                    |
| <b>Water Solubility</b>                        | 2.7 g/100ml (20°C)             |                                          |
| <b>Solubility in other solvents</b>            | No information available       |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                |                                          |
| <b>Component</b>                               | <b>log Pow</b>                 |                                          |
| Boron,                                         | 0.64                           |                                          |
| trihydro(2-methyl-2-propanamine)-,             |                                |                                          |
| (T-4)-                                         |                                |                                          |
| tert-Butylamine                                | 0.4                            |                                          |
| <b>Vapor Pressure</b>                          | No information available       |                                          |
| <b>Density / Specific Gravity</b>              | No data available              |                                          |
| <b>Bulk Density</b>                            | No data available              |                                          |
| <b>Vapor Density</b>                           | Not applicable                 | Solid                                    |
| <b>Particle characteristics</b>                | No data available              |                                          |

### 9.2. Other information

|                          |                                                                                    |
|--------------------------|------------------------------------------------------------------------------------|
| <b>Molecular Formula</b> | C4 H14 B N                                                                         |
| <b>Molecular Weight</b>  | 86.97                                                                              |
| <b>Flammable solids</b>  | Burning rate or burning time = > 2.2 mm/s or < 45 secs<br>Wetted zone passed - Yes |
| <b>Evaporation Rate</b>  | Not applicable - Solid                                                             |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

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## 10.2. Chemical stability

Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

## 10.4. Conditions to avoid

Temperatures above 65°C. Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.

## 10.5. Incompatible materials

Acids. Finely powdered metals. oxygen.

## 10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen. Oxides of boron.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 4 |

| Component                                        | LD50 Oral               | LD50 Dermal | LC50 Inhalation             |
|--------------------------------------------------|-------------------------|-------------|-----------------------------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | LD50 = 96 mg/kg ( Rat ) | -           | -                           |
| tert-Butylamine                                  | 464 mg/kg ( Rat )       | -           | LC50 = 3.8 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

#### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.



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(j) aspiration hazard; Not applicable  
Solid

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Fish                        | Water Flea                        | Freshwater Algae |
|-----------------|----------------------------------------|-----------------------------------|------------------|
| tert-Butylamine | Salmo Gairdneri: LC50: 28-270 mg/L/96h | Daphnia magna: EC50: 136 mg/L/24h |                  |

| Component       | Microtox                               | M-Factor |
|-----------------|----------------------------------------|----------|
| tert-Butylamine | Pseudomonas putida: EC50: 110 mg/L/96h |          |

### 12.2. Persistence and degradability

#### Persistence

#### Degradation in sewage treatment plant

Persistence is unlikely.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component                                        | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------------|---------|-------------------------------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | 0.64    | No data available             |
| tert-Butylamine                                  | 0.4     | No data available             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors.

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance.  
This product does not contain any known or suspected substance.

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## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                   |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment. |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a>                                 |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                                         |
|-----------------------------------------|-----------------------------------------|
| <b>14.1. UN number</b>                  | UN2926                                  |
| <b>14.2. UN proper shipping name</b>    | Flammable solid, toxic, organic, n.o.s. |
| <b>Technical Shipping Name</b>          | Borane-tert-butylamine complex          |
| <b>14.3. Transport hazard class(es)</b> | 4.1                                     |
| <b>Subsidiary Hazard Class</b>          | 6.1                                     |
| <b>14.4. Packing group</b>              | II                                      |

### ADR

|                                         |                                         |
|-----------------------------------------|-----------------------------------------|
| <b>14.1. UN number</b>                  | UN2926                                  |
| <b>14.2. UN proper shipping name</b>    | Flammable solid, toxic, organic, n.o.s. |
| <b>Technical Shipping Name</b>          | Borane-tert-butylamine complex          |
| <b>14.3. Transport hazard class(es)</b> | 4.1                                     |
| <b>Subsidiary Hazard Class</b>          | 6.1                                     |
| <b>14.4. Packing group</b>              | II                                      |

### IATA

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
| <b>14.1. UN number</b>                  | UN2926                                   |
| <b>14.2. UN proper shipping name</b>    | FLAMMABLE SOLID, TOXIC, ORGANIC, N.O.S.* |
| <b>Technical Shipping Name</b>          | Borane-tert-butylamine complex           |
| <b>14.3. Transport hazard class(es)</b> | 4.1                                      |
| <b>Subsidiary Hazard Class</b>          | 6.1                                      |
| <b>14.4. Packing group</b>              | II                                       |

|                                    |                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.5. Environmental hazards</b> | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|------------------------------------|----------------------------------------------------------------------------------------------------------|

|                                           |                                  |
|-------------------------------------------|----------------------------------|
| <b>14.6. Special precautions for user</b> | No special precautions required. |
|-------------------------------------------|----------------------------------|

|                                                                      |                                |
|----------------------------------------------------------------------|--------------------------------|
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods |
|----------------------------------------------------------------------|--------------------------------|

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Borane-tert-butylamine complex

Revision Date 25-Jan-2024

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                        | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | 7337-45-3 | 230-851-5 | -      | -   | X     | X    | KE-04194 | -    | -    |
| tert-Butylamine                                  | 75-64-9   | 200-888-1 | -      | -   | X     | X    | KE-24864 | X    | X    |

| Component                                        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------------------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | 7337-45-3 | X    | ACTIVE                                        | X   | -    | X    | -     | X     |
| tert-Butylamine                                  | 75-64-9   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

Not applicable

| Component                                        | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | 7337-45-3 | -                                                                   | -                                                                             | -                                                                                                     |
| tert-Butylamine                                  | 75-64-9   | -                                                                   | -                                                                             | -                                                                                                     |

#### Seveso III Directive (2012/18/EC)

| Component                                        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Boron, trihydro(2-methyl-2-propanamine)-, (T-4)- | 7337-45-3 | Not applicable                                                                            | Not applicable                                                                           |
| tert-Butylamine                                  | 75-64-9   | Not applicable                                                                            | Not applicable                                                                           |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

#### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

#### National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

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**WGK Classification** See table for values

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                              |
|-----------------|---------------------------------------|------------------------------------------------------|
| tert-Butylamine | WGK2                                  | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |

| Component       | France - INRS (Tables of occupational diseases)               |
|-----------------|---------------------------------------------------------------|
| tert-Butylamine | Tableaux des maladies professionnelles (TMP) - RG 49,RG 49bis |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H228 - Flammable solid  
H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H332 - Harmful if inhaled  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H411 - Toxic to aquatic life with long lasting effects  
H225 - Highly flammable liquid and vapor  
H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
H412 - Harmful to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

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Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department        |
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| <b>Revision Date</b>    | 25-Jan-2024                                        |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**